

OriginX: Measuring One-Shot Recovery in a Frozen Robot Policy

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Abstract

Can one image-conditioned instruction recover a frozen robot policy’s failure within its original action budget? We study all 1,004 failures from a completed 2,500-episode RoboCasa365 evaluation using control and assisted replays. Verification separates comparable assistance from task recovery: all 869 eligible pairs produce different first-intervention action outputs, but only 56 succeed. The full cohort retains 56 rescues, 813 non-rescues, 28 unknowns and 107 initial-state deviations, yielding 5.58% confirmed recovery. A separate, post-start scope-amended follow-up covers 50 tasks with one fresh case each and 350 arm outcomes, adding simple text controls, vision-blind advice and the original base policy. B’s unchanged and image-conditioned arms both achieve 29/50 raw successes. Their 44 matched pairs contain one assisted-only success; initial deviations leave full-cohort net-effect bounds of $[-8, +4]$ percentage points. Image-conditioned versus vision-blind assistance has two treatment-only successes among 42 matched pairs, but one uses unchanged-instruction fallback rather than visual advice. Its bounds are $[-6, +16]$ points. The studies identify a gap between changing policy outputs and reliably improving task outcomes, while exposing delivery, replay and latency limitations. They establish neither overall benchmark improvement nor a unique visual-reasoning advantage.

1. Introduction

A robot can understand a household instruction yet fail to execute it. After a missed interaction or an unproductive continuation, a vision-language assistant could describe what to do next while a frozen vision-language-action (VLA) policy continues controlling the robot. This interface is appealing because it requires neither policy retraining nor a new motor-action representation. Its value depends on a less obvious question: *does a changed instruction merely change the action output, or does it recover the task?*

Language-mediated planning and recovery are well established [1–5]. Here we study a restricted setting that exposes both the opportunity and the limitations of this

interface: one image-conditioned instruction, a fixed mid-point trigger, an unchanged robot policy, and the original environment-step budget. The intervention occurs during a fresh replay of a historically failed episode; it does not restart the robot from that episode’s terminal failed state. Neither a visible failure nor a hidden simulator failure predicate is required to trigger assistance.

OriginX is a frozen derivative of Xiaomi-Robotics-1-RoboCasa365 [6]. Its completed 50-task evaluation yields 1,496 successes and 1,004 failures. We retain that historical record and attempt one control and one assisted replay of every failure. Three research questions organize the study: **RQ1:** How much of this cohort admits a verifiable assisted comparison? **RQ2:** When advice changes the policy output, how often and how quickly does the task recover? **RQ3:** Where does recovery remain limited, and what overhead does assistance introduce?

These questions require distinguishing a retry from a comparable intervention. A reset may produce a different initial scene; paired arms may diverge before the instruction changes; generated advice may never reach the policy. Our offline verifier compares historical initial fingerprints, paired preintervention traces, and linked evidence of actual response application. The assistant itself sees only current RGB views, the task instruction, and its remaining action budget. Every historical failure retains a terminal category, including cases whose comparison cannot be validated.

The evidence reveals a gap between output sensitivity and successful recovery. All 869 eligible pairs receive verified advice and produce a different action hash at the first intervention query, where their state and images still match. Only 56 finish the task. Valid-assistance coverage is 86.55%; multiplying it by the 6.44% eligible-pair rescue fraction gives the fixed-cohort yield of 5.58%. Composite-Unseen has the highest coverage but the lowest conditional rescue fraction. Successful cases often finish well after intervention: 29 of 56 finish only after using more than half the remaining budget. Assistance also introduces a median 191-second request wait.

Our contributions are an executed, fully accounted study of 1,004 historical failures; a replay-verification protocol that separates coverage, action-output change, and task recovery; and trace-grounded analyses of recovery timing, task heterogeneity, and inference overhead. The novelty is

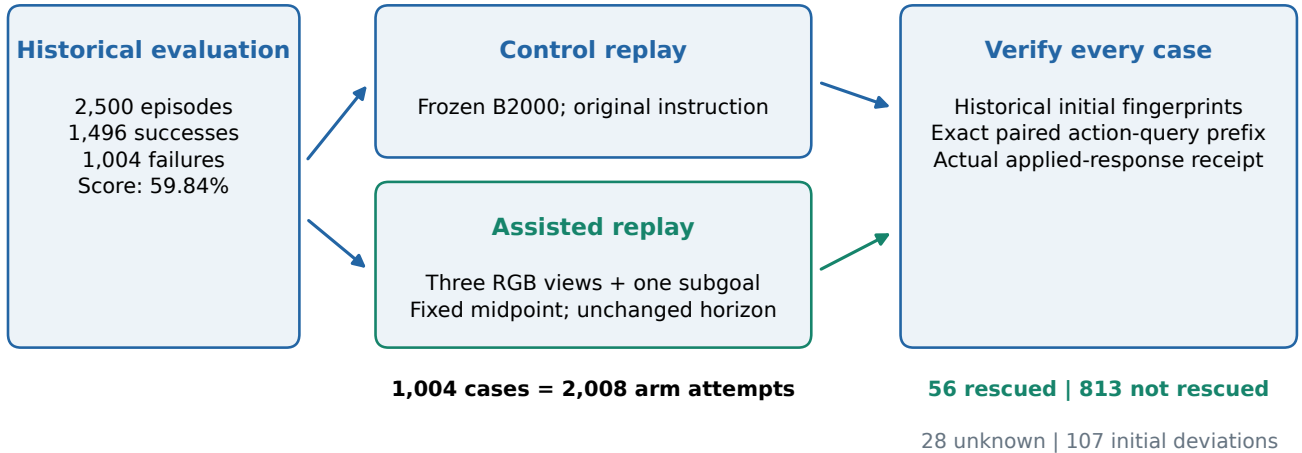


Figure 1. **Two records, one fixed failure cohort.** The original evaluation remains 1,496/2,500. Each of its 1,004 failures is assigned one control and one assisted replay. Assistance changes only the instruction at a fixed query; the robot weights and action horizon remain frozen. Offline replay checks and actual-response evidence determine whether the pair is eligible. All four terminal categories stay in the cohort denominator.

the empirical characterization under these constraints, not language correction itself. We preserve the 813 eligible non-rescues and 135 unknown or deviating cases. The historical study omits successful episodes and simple language controls; Section 6 adds a separate, smaller control study. Neither supports an unrestricted deployment-wide improvement or an Astra-specific reasoning claim.

2. Related Work

Language, feedback, and recovery. SayCan grounds language plans in robot affordances [1]; Inner Monologue adds environmental feedback [2]. REFLECT, RACER, and FLARE develop explanation, correction, and failure-aware recovery [3–5]. RT-H learns language-action hierarchies [14]; CycleVLA uses progress-aware backtracking [15]; ReCoVLA learns VLM-guided residual recovery [16]. Language recovery is therefore not a new interface. Our study measures its limits with a frozen policy, one scheduled instruction, and the original action horizon. We do not reproduce or claim superiority to these systems.

Policy and evaluation substrate. OriginX adapts Xiaomi-Robotics-1 [6] using Action LoRA [7] and observation-conditioned modulation related to adaptive-normalization transformers [8]; its released flow objective is inherited [9]. The assistance study freezes this substrate and does not isolate its branch’s contribution. RoboCasa and RoboCasa365 provide household tasks and the evaluation setting [10–12]. SIMPLER emphasizes specification of simulation interfaces [13], while LIBERO-RECOVER

constructs a dedicated failure-recovery benchmark [17]. Our contribution is within-study accounting: recorded initial observations, paired prefixes, and actual advice application distinguish a successful retry from a verified matched rescue.

3. Frozen Policy and Assistance Interface

3.1. The OriginX policy substrate

The frozen endpoint $\pi_B = \pi_{\theta_0, \alpha, \phi}$ contains the released Xiaomi parameters θ_0 , an Action LoRA adapter α , and a 4,216,839-parameter observation-conditioned branch ϕ that modulates the action transformer’s adaptive normalization. Its complete architecture, data lineage, and training losses are documented in the supplement. The assistance experiment updates none of these parameters and supplies no privileged stage labels at inference.

The public adapter alias A2000 refers to unchanged weights trained for 1,613 updates; B2000 identifies the branch after 2,000 updates. This policy substrate is not a separately validated methodological contribution. An earlier 600-pair comparison against the original policy produced 419 versus 413 successes, with 53 B-only and 47 original-only outcomes (exact McNemar $p = 0.6173$). It changes both adapter and branch under a different protocol and neither isolates the branch nor supplies a matched baseline for the later 2,500 episodes.

3.2. A single scheduled language intervention

The policy replans every 16 environment steps. For horizon H , the assistance trigger is

$$\tau = 16 \lceil H/32 \rceil. \quad (1)$$

This is the first regular query at or after half the horizon. The trigger depends only on the fixed action budget, not on hidden success state, retrospective failure information, or a learned stall score. The original failure list selects the *study cohort*; its outcome labels are not supplied to the assistant.

At τ , the wrapper sends the original task instruction, current left/right/wrist RGB images (256×256 each), and remaining environment steps to `gpt-6-astra` at high reasoning effort. Routing identifiers associate each response with its case. Hidden simulator state, success predicates, future frames, and original outcome labels are excluded. The robot policy continues receiving its ordinary proprioceptive observations; these are not additionally sent to Astra.

The supplement reproduces the fixed prompt, response-schema factory, and append rule. The response provides at most 1,500 characters of subgoal text. The wrapper preserves the original instruction, appends “Immediate next actions:” and that text, then appends “Then complete the original task.” The augmented instruction persists for subsequent policy queries. Astra receives no second observation and produces no motor actions. Its generated rationale is not a verified explanation of failure.

The broker groups at most eight distinct requests into labeled image contact sheets. Each response must pass the recorded schema and identity checks. The prompt forbids tool use, and event traces are checked for tool calls. The CLI invocation records the requested model and reasoning effort; it is not an independently attested identity of service-side model weights. Batch context is a shared resource and a possible source of dependence between cases, not eight separate model invocations.

3.3. Budget and runtime invariants

Both arms preserve seed, task, native reset, endpoint, observation history, five Euler integration steps, replanning interval, crop ratio, and registered horizon. Simulation pauses while awaiting assistance, for at most 1,200 wall-clock seconds. This wait consumes additional inference and wall time but does not add action steps. The control is therefore *action-budget matched*, not compute- or latency-matched. Read-only RNG keepalives keep the same policy socket alive without reconnecting or resetting it. Missing, invalid, or unapplied responses remain technical outcomes; completed failures are not rerun to select a better outcome.

4. Paired Replay Design

4.1. Historical record and fixed cohort

The historical native-reset evaluation contains 50 tasks with 50 episodes each: 18 Atomic-Seen, 16 Composite-Seen, and 16 Composite-Unseen, in pretraining scenes. Environment and policy seeds are fixed in [2090900000, 2090902499]; task horizons follow RoboCasa 1.0.1. All 2,500 planned episodes have terminal records, with zero missing or infrastructure-unknown outcomes. Every failure reaches its original horizon. The original Counter reset implementation is checked before and after each episode.

We define the rescue cohort before replay as

$$\mathcal{F} = \{i : Y_i^{\text{historical}} = 0\}, \quad |\mathcal{F}| = 1,004. \quad (2)$$

It spans 49 tasks; CloseFridge contributes no cases because its historical 50 episodes all succeeded. Each case is assigned two arms, giving 2,008 planned arm attempts but only 1,004 unique cases. The first campaign permanently claims 36 cases; an infrastructure continuation claims the other 968. This partition is fixed at whole-case level before continuation. No later outcome is substituted for a first-partition outcome.

Four infrastructure-development cases, indices 48, 97, 249, and 345, were run separately before the main study and yielded zero pilot rescues. Their pilot outcomes are excluded, but their identities remain in the primary cohort and are explicitly development-exposed. We report a leave-these-identities-out sensitivity analysis rather than describe the full cohort as untouched test data.

4.2. Verification gates and terminal categories

For each case, the verifier checks three distinct conditions. **Initial agreement** requires both replay initial fingerprints to agree with the historical case, including the recorded scene, observation, instruction, and state evidence. **Prefix agreement** requires exact paired state/image/instruction/action-query hashes before the intervention. **Applied assistance** requires linked request, image, response, CLI, and policy-query evidence showing that a valid Astra response actually reached the assisted arm. The state checks are offline measurements; their privileged contents are never policy or assistant inputs.

Let M_i indicate valid completed arms with matching historical initial fingerprints and paired preintervention evidence. Let A_i indicate verified applied assistance and $E_i = M_i A_i$. The rescue indicator is

$$R_i = E_i \mathbf{1}[Y_i^C = 0, Y_i^A = 1]. \quad (3)$$

Here C and A denote the control and assisted outcomes. A valid response with no outcome, or an assisted success

without matching evidence, is not a strict rescue. An eligible pair that does not satisfy the outcome transition is a non-rescue. Confirmed initial mismatches receive their own deviation category; otherwise missing or invalid evidence remains unknown. These categories are exhaustive and mutually exclusive in the settled ledger.

A further trace audit checks equal state and RGB hashes at the intervention query itself, before the changed instruction is evaluated. The checks strengthen a within-case comparison, but do not prove simulator equivalence outside the recorded contracts. In particular, preintervention traces are compared between the two new arms; they are not reconstructed historical trajectories. The historical archive contains outcome records and initial fingerprints, not videos of the original failures. We cannot retrospectively identify the original first physical error from a hash.

4.3. Coverage and recovery are different quantities

The primary estimand is the realized confirmed rescue fraction of the fixed historical cohort, $\hat{r}_{\mathcal{F}} = \sum_i R_i/N$, with $N = 1004$. Its exact accounting identity is

$$\underbrace{\frac{R}{N}}_{\text{confirmed yield}} = \underbrace{\frac{E}{N}}_{\text{valid coverage}} \underbrace{\frac{R}{E}}_{\text{conditional recovery}}, \quad E > 0. \quad (4)$$

This decomposition distinguishes a missing valid comparison from an observed failure after valid assistance. Unknowns and deviations remain in N . The matched-complete count M is an intermediate evidence quantity; R/M is reported in the supplement. Neither R/E nor R/N is an overall deployment success rate. The 1,496 historical successes were not retested, so assistance-induced regressions on them are unmeasured.

The counts form a census of a failure-selected cohort. Task, timing, and batch analyses are retrospective, not pre-registered tests. We report technical attrition directly, without assuming missing-at-random outcomes or imputing success. Zero control successes are an observed property of the eligible replays, not an eligibility requirement. The paired asymmetry does not establish full-population improvement.

5. Results

5.1. Baseline performance and the opportunity set

Table 1 records the frozen endpoint. The 49.25-point Atomic-Seen versus Composite-Unseen gap motivates studying failures, but does not diagnose their causes. Composite-Unseen supplies 530 of the 1,004 failures (52.79%). GatherTableware, HeatKebabSandwich, and PanTransfer each fail all 50 historical episodes. These failures identify an opportunity set for investigation; they do not establish that an instruction-level correction is sufficient.

Historical split	Tasks	Successes	Rate
Atomic-Seen	18	747/900	83.00%
Composite-Seen	16	479/800	59.88%
Composite-Unseen	16	270/800	33.75%
All tasks	50	1496/2500	59.84%

Table 1. **Frozen historical evaluation.** This single-arm record defines the later failure cohort. It is unchanged by assisted replays. Equal trials per task make the pooled rate equal to the task mean.

Terminal category	Cases	Fraction of cohort
Strict rescue	56	5.58%
Eligible non-rescue	813	80.98%
Technical unknown	28	2.79%
Initial-state deviation	107	10.66%
Fixed cohort	1004	100.00%

Table 2. **Complete accounting, incomplete valid coverage.** All planned attempts are terminal. Unknowns and deviations are not converted to policy failures or removed from the denominator; rounded percentages need not sum exactly.

We omit unmatched leaderboard comparisons because different execution stacks and unavailable paired outcomes prevent attribution of small score differences.

5.2. RQ1: How much of the cohort is verifiable?

Table 2 gives the complete ledger. We obtain $M = 874$ matched completed replay pairs and $E = 869$ pairs with verified applied assistance. Thus $86.55\% \times 6.44\% = 5.58\%$ up to rounding: the conditional result alone omits the 13.45% without a valid intervention comparison. Complete accounting does not mean complete valid coverage.

Among eligible pairs, the complete paired table is 56 assisted-only successes and 813 failures in both arms, with zero control-only and zero both-arm successes. The 874-pair raw matched table adds five both-failure pairs whose assistance evidence is invalid; those five remain unknown, not eligible non-rescues. Their presence illustrates why terminal success bits alone cannot establish that an intervention was tested.

The first partition yields 0 rescues, 3 non-rescues, 28 unknowns, and 5 initial deviations. The continuation yields 56, 810, 0, and 102 respectively. All 28 unknowns therefore arise in the first partition. The continuation changed infrastructure handling, including same-socket keepalives and continued dispatch after validated native-reset deviations. We retain partition membership because this operational history can affect technical coverage. The study is not an uninterrupted homogeneous execution, although its endpoint, action budget, and intervention rule are fixed.

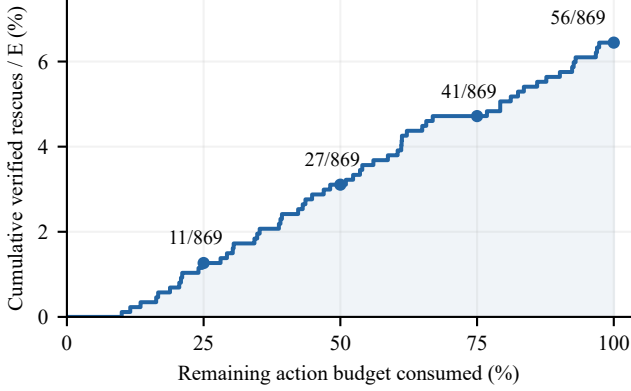


Figure 2. **Recovery takes continued action.** At 25/50/75/100% of the remaining budget, 11/27/41/56 cases have succeeded, out of 869 eligible pairs. All 813 eligible non-rescues reach their original horizons. The curve describes recorded success timing, not performance under alternative intervention times.

5.3. RQ2: Does advice change actions and outcomes?

At the intervention query, all 869 eligible pairs have identical recorded state and RGB hashes, different instruction hashes, and immediately different returned action hashes. The action hash covers the returned array’s shape, dtype, and bytes; it does not include the instruction. Thus zero-step action-output divergence is observed in all 56 rescued and all 813 non-rescued pairs, with no missing trigger record. This establishes a computational difference, not its magnitude or usefulness. Hashes cannot distinguish a meaningful motor correction from a tiny numerical perturbation, and this comparison does not independently randomize service scheduling or arithmetic effects.

To measure recovery timing, define $u_i = (T_i - \tau_i) / (H_i - \tau_i)$ for a rescued pair, where T_i is its recorded success step. Figure 2 plots

$$G_E(u) = \frac{1}{869} \sum_{i: R_i=1} \mathbf{1}[u_i \leq u], \quad (5)$$

with zero contribution from non-rescues. Unlike a CDF conditioned only on success, this curve retains all eligible failures in its denominator and ends at 6.44%, not 100%.

Among the 56 rescues, median u_i is 51.67% (interquartile range 30.45–77.41%), and the median continuation is 361 environment steps. Fifteen rescues finish in the last quarter of their remaining budget; seven use more than 90%. A short observation window after advice would therefore miss recorded successes. An earlier trigger might help, harm, or leave outcomes unchanged; that counterfactual has not been tested.

5.4. RQ3: Where does recovery remain limited?

Table 3 and Fig. 3 expose two different limitations. Valid coverage is 89.54%, 75.70%, and 92.26% in Atomic-Seen, Composite-Seen, and Composite-Unseen; conditional recovery is 8.76%, 10.29%, and 3.89%. The lowest yield in Composite-Unseen is therefore not the arithmetic consequence of lower coverage. Conversely, Composite-Seen loses 70/321 cases to initial-state deviations, compared with 21/530 in Composite-Unseen. These patterns separate the accounting factors without identifying why the splits differ.

Rescues span 27 of the 49 tasks with historical failures. BreadSelection and PrepareCoffee account for 13/56 rescues (23.21%); the five largest rescue counts contribute 24/56 (42.86%). Equal weighting of failure-containing tasks gives 7.83%, compared with the pooled 5.58%; these weight different task mixtures. Complete rows are retained in the supplement.

The three originally zero-success tasks yield one rescue in 150 failures (135 eligible pairs); the five lowest historical task scores yield one in 237 (219 eligible). These difficult subsets remain unresolved. However, the full task scatter does not support a monotonic difficulty law: historical task success has descriptive Spearman correlation 0.075 with fixed-cohort yield (49 tasks), and 0.117 with eligible-pair recovery (48 tasks). The bottom-five finding is a retrospective subset description, not a general predictor of recoverability.

Whole-task resampling within original splits gives a 3.59–7.95% percentile range for confirmed yield. This exploratory task-mixture sensitivity is not a confidence interval for the finite census or a causal effect; its assumptions and the separate batch sensitivity are detailed in the supplement.

Development exposure. Removing all four disclosed pilot identities yields 55/1,000 confirmed rescues (5.50%); this sensitivity does not replace the primary cohort. The separate pilot’s 0/4 outcomes are never added to it.

5.5. Two matched outcomes within the same task

Figure 4 shows the complete two-case historical OpenDrawer failure subset. In case 186 (seed 2090900186), As-tra supplies a subgoal about approaching the visible drawer handle, grasping it, and pulling outward. The applied response is linked to the policy’s changed instruction. Its assisted arm succeeds at step 674, while the unassisted control exhausts the 750-step horizon. Case 586 (seed 2090900586) also receives valid advice but neither arm succeeds. Equal initial and paired-prefix evidence localizes the experimental difference to the intervention period under the checked contracts. It does not tell us which phrase, visual interpretation, or subsequent physical motion produced the positive outcome; the recorded trigger images are not a postintervention trajectory.

Original failure split	Fixed cases	Eligible E	Rescued	Unknown	Initial deviation	Rescued/fixed cases
Atomic-Seen	153	137	12	0	16	7.84%
Composite-Seen	321	243	25	8	70	7.79%
Composite-Unseen	530	489	19	20	21	3.58%
All	1004	869	56	28	107	5.58%

Table 3. **Task familiarity and technical coverage.** Counts are conditioned on historical failure. The lower confirmed rescue fraction in Composite-Unseen is descriptive: the splits have different tasks and horizons, and no controlled generalization effect is identified. In particular, Composite-Seen has a much higher initial-deviation count.

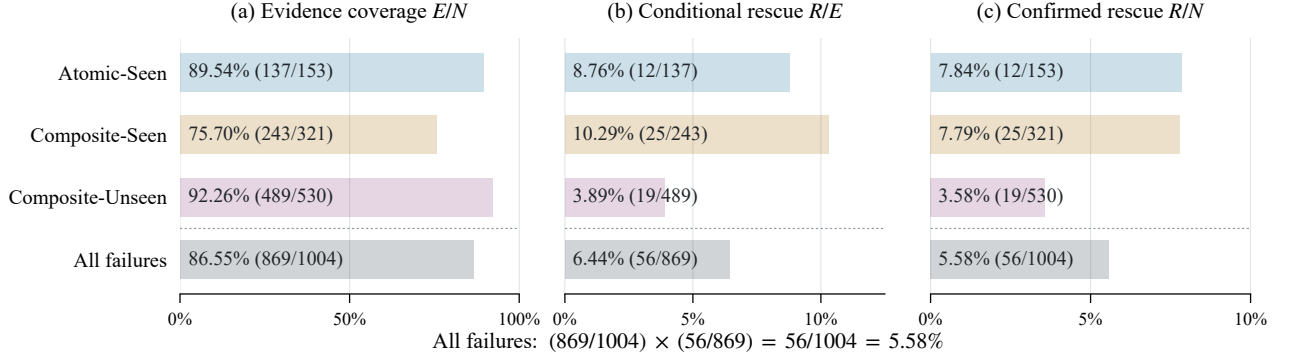


Figure 3. **Technical coverage and behavioral recovery separate.** Each row factorizes the confirmed yield under Eq. 4. Composite-Unseen has the highest eligible coverage and the lowest conditional rescue fraction. This is a descriptive split comparison, not a controlled effect of task novelty.

Primary-study accounting	Recorded value
Unique CLI invocations	135 (134 successful)
Known input tokens	2,592,713
Known output tokens	195,237
Reported reasoning tokens	75,511
Median CLI time / invocation	46.16 s
Median eligible-case request wait	191.38 s
95th percentile eligible-case wait	287.39 s

Table 4. **Actual inference overhead.** Invocations are deduplicated across batches and receipts. One error lacks counters; complete token totals and dollar cost are unknown. Reasoning tokens are not added again to output tokens. Request waiting includes queue/transport time and is distinct from CLI runtime.

5.6. Operational cost of the intervention

Table 4 distinguishes request waiting from CLI runtime. Total known invocation latency is 6,186.44 s; batching prevents interpreting this as per-case latency. Eligible-case waits are available for 869 cases, excluding timed-out and invalid cases. The successful OpenDrawer case waits 327.17 s while its batch executes for 48.98 s. The separate pilot adds 19,071 input and 629 output tokens (137 reasoning tokens) and 24.35 s of CLI runtime. Failed and unap-

plied responses remain in accounting. Verified monetary cost is unavailable.

6. Scope-Amended 50-Case Follow-Up

Question and sampling. To examine simple-language explanations and potential regressions, a separate study originally sealed 2,500 fresh cases (50 per task). After execution began, the user requested a three-hour delivery window at 20:46:19 UTC on October 9, 2026. A deterministic amendment retains the minimum originally assigned fresh seed for each task: 50 cases and 350 arm assignments. Selection reads task/seed fields, not outcomes, but occurs after launch; this is a scope-amended follow-up, not completion of the original confirmation plan. The 191 already claimed out-of-cohort arm records remain archived separately. Historical results and their denominators are unchanged.

Interventions and policy comparison. The frozen B endpoint receives five arms: unchanged-instruction sham (C), original-task restatement (R), fixed generic recovery text (G), vision-blind Astra advice (L), and image-conditioned Astra advice (V). The original Xiaomi base receives C/V on the same cases. All use Eq. 1, their frozen horizons, and the existing append rule; L omits RGB. C traverses the wrap-

(a) Case 186: rescued

Open the left drawer.



Advice: "Lower your view and gripper below the countertop to the left drawer."

Control: fails at 750; assisted: succeeds at 674

Both: intervention at 384 / 750 steps; initial state and 24-query prefix match.

(b) Case 586: not rescued

Open the right drawer.



Advice: "Back the wrist slightly away from the cabinet face, align the gripper with the horizontal handle on the right drawer directly below the countertop, grasp the handle, and pull straight outward."

Control: fails at 750; assisted: fails at 750

Figure 4. **Actual intervention observations for both historical OpenDrawer failures.** The rows show the original three RGB views at the trigger, not reconstructed videos or generated scenes. Both pairs match their historical initial fingerprints and have 24 exact paired preintervention queries; both trigger at step 384 with horizon 750. Case 186 is rescued (assisted success at 674; control fails at 750). Case 586 is not (both fail at 750). These observations and instructions illustrate two outcomes of the same assistance interface; they do not establish a physical causal explanation.

per without changing instruction bytes. The base is closely related to B; it is not cross-architecture replication. Formal GR00T evaluation is unrun, and its separate development checks provide no transfer result.

Estimand and evidence requirements. The primary information contrast is B:V-L; C comparisons quantify gains and harms against no instruction change, and R/G address simpler text. Each paired effect is $(W - L)/50$, where W is treatment-only success and L reference-only success. Initial fingerprints, exact pretrigger traces, and trigger observations must agree. Valid same-prefix early termination remains a paired result without requiring assistant use; clean delivery failure continues the original instruction. Infrastructure failures, initial deviations, and prefix deviations are retained separately. Missing endpoints produce identification bounds, not a fabricated complete-cohort rate. Task-block intervals are conditional descriptive sensitivity, not a multiple-comparison superiority claim.

Results and delivery. All 350 assignments have valid terminal outcomes, but this does not mean 350 valid interventions. Raw B successes are C/R/G/L/V = 29/28/27/27/29 out of 50: V gives no raw gain over C. B:V-C has 44 matched pairs, one V-only success and no C-only success, with bounds of $[-8, +4]$ points. For B:V-L, 42 pairs match, with two V-only successes and bounds of $[-6, +16]$ points. All unresolved comparisons in Table 5 are initial deviations; technical-unknown and prefix-deviation counts are

Contrast	M	W	L	U	Δ bounds (pp)
B: R-C	47	0	1	3	$[-8.0, -2.0]$
B: G-C	44	1	1	6	$[-12.0, +0.0]$
B: L-C	45	0	1	5	$[-12.0, -2.0]$
B: V-C	44	1	0	6	$[-8.0, +4.0]$
B: V-L	42	2	0	8	$[-6.0, +16.0]$
Base: V-C	47	0	1	3	$[-6.0, +0.0]$

Table 5. **Separate 50-case follow-up.** M : matched valid pairs; W/L : treatment/reference-only successes; $U = 50 - M$: unresolved comparisons, including deviations. Bounds retain known one-sided endpoints. All counts come from the final reconciled report.

zero. Raw rate differences and matched effects are therefore distinct summaries.

Only 20 B:V assignments receive applied visual advice; 20 terminate early and 10 use clean unchanged-instruction fallback. One of the two V-L wins is such a fallback: Pick-PlaceCounterToStove succeeds in C and V at step 574 while applied L advice fails. The other, WashFruitColander, receives V advice and succeeds at 2,347 steps, while C/L fail at 3,150. Thus two V-arm wins do not mean two rescues by visual advice. The base's raw count falls from 26 to 25; its matched comparison has one regression and no gain. Neither policy provides evidence of an overall assistance benefit here.

Scope. One seed per task cannot estimate within-task variation. Administrative truncation, inherited claims and infrastructure continuations are disclosed in the supplement; the 191 excluded prior arms remain archived. The small cohort, incomplete paired comparability and delivery fallbacks preclude broad transfer or superiority claims. Inference accounting includes 72 unique CLI invocations with complete input/output totals; the supplement separates these costs from task outcomes.

7. Interpretation and Limitations

Sensitivity is not successful control. The action audit shows unequal first-query outputs in every eligible pair. It does not by itself establish semantic use of advice or exclude numerical and service effects; the historical study has no same-wrapper sham control. The smaller follow-up addresses this limitation within a separate cohort. A hash may differ because of a small numerical change. The 56 successful paired outcomes provide stronger task-level evidence, while 813 verified applications remain unsuccessful. Accordingly, output differences should not be treated as a demonstrated recovery mechanism without outcome and trajectory validation.

Attribution to the interface, not a unique reasoning mechanism. The historical contrast changes both instruction and external inference. The follow-up adds restatement, fixed generic text, and vision-blind advice, but only one seed per task. It is not a compute-matched comparison, has no alternative assistant, and cannot establish an Astra-specific reasoning mechanism. Nor does it identify the contribution of OriginX’s continuous branch. The earlier unsuccessful paired confirmation and absent A-only ablations remain part of the evidence, not objections that can be resolved by a stronger aggregate claim.

Selected cases and replay attrition. The cohort is postselected on historical failure. Successes were not revisited, four case identities were used during infrastructure development, and unmatched or invalid pairs comprise 13.45% of the cohort. Assistance cannot be declared beneficial overall without evaluating a frozen rule on a fresh complete manifest, including episodes it might harm. Exact recorded prefixes help isolate the within-pair intervention period but do not remove selection, possible batch dependence, simulator limitations, or service-version drift. A blinded independent rerun is still needed.

Reproducibility and scope. The evidence includes model/configuration identities, case manifests, per-arm outcomes, trace hashes, actual requests/responses, source snapshots, and deduplicated usage. The simulator uses OSMesa

with disclosed rendering and context-management changes; query-pixel checks support internal consistency, not universal simulator equivalence. The release’s portable loader has CPU contract checks but has not undergone a new real-model GPU parity run. The historical results cover one adapted policy. The follow-up adds its closely related original base, not an independent architecture. They do not establish real-robot safety, cross-embodiment transfer, or SIMPLER performance. The supplement separates tested contracts from untested portability claims.

8. Future Directions

The first priority is an independently sealed, adequately powered replication of the language contrasts, including successes that advice may harm. One seed per task cannot estimate within-task variance; multiple fresh seeds and fixed task-block analysis should test whether image-conditioned advice improves net outcomes over vision-blind and simple text controls. An alternative assistant and an actually evaluated different policy architecture are necessary before attributing gains to Astra or claiming transfer.

Timing and feedback should be varied separately from instruction content: compare a predeclared trigger grid, then an observation-only trigger developed on disjoint data. Closed-loop advice must share total token, wall-time, and action limits with its comparator. Recovery trajectories can support distillation only after postintervention state-action sequences are retained and validated, with failures preserved and new evaluation seeds sealed. These are testable extensions, not achieved results; physical deployment additionally needs contact, stopping, and latency safeguards.

9. Conclusion

Matched replay produces different first-intervention action outputs in all 869 valid comparisons, but only 56 task rescues. Fixed-cohort accounting yields 56/1,004 confirmed rescues, retaining 813 non-rescues, 28 unknowns, and 107 initial-state deviations. Coverage, behavioral recovery, continuation budget, and wall-clock latency expose distinct limits of this frozen-policy interface. The original 59.84% evaluation remains unchanged. The 50-case follow-up completes its 350 assignments but shows no raw B success-rate gain (29/50 in C and V); unmatched initial states keep the paired full-cohort effect unresolved. A V-L win obtained through fallback further cautions against attributing system-level differences to visual reasoning. Reliability and net benefit remain open questions.

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